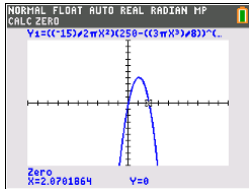
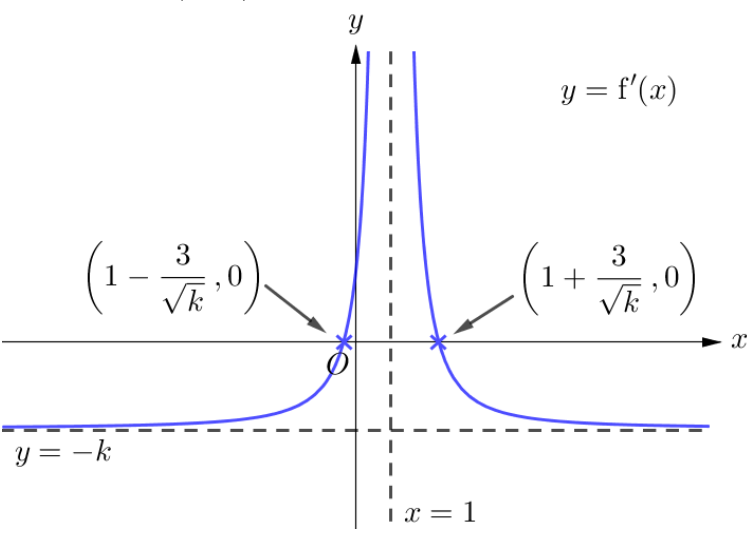
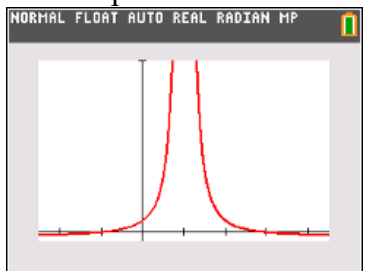
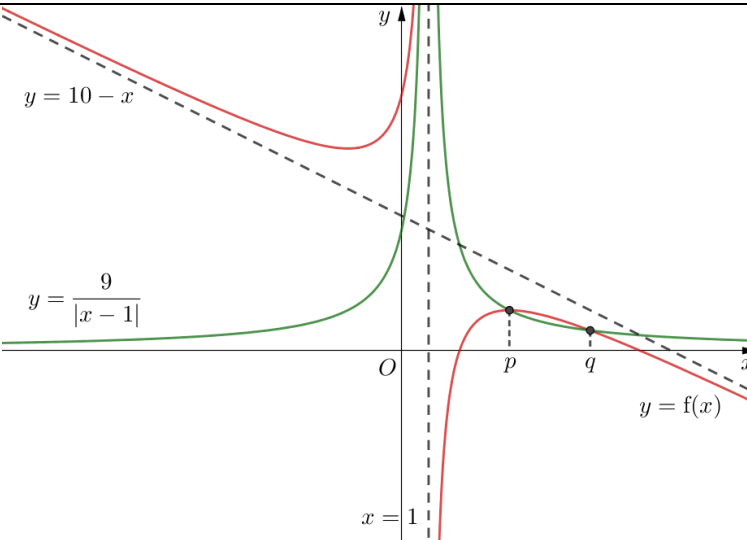


Q1	Suggested Solutions	Marking Scheme
	The first 3 terms of the GP are $a + 5d, a + 3d, a + 2d$. common ratio, $r = \frac{a + 3d}{a + 5d} = \frac{a + 2d}{a + 3d}$ $(a + 3d)^2 = (a + 2d)(a + 5d)$ $a^2 + 6ad + 9d^2 = a^2 + 7ad + 10d^2$ $d^2 + ad = 0$ $d = -a \text{ (Since } d \neq 0\text{)}$ $r = \frac{a + 3(-a)}{a + 5(-a)}$ $= \frac{1}{2}$	M1: Equating ratio (r or $1/r$) M1: Find value of r using value of d found
	Required sum $= T_{15} + T_{16} + \dots$ $= (a + 5d)r^{14} + (a + 5d)r^{15} + \dots$ $= \frac{(-4a)\left(\frac{1}{2}\right)^{14}}{1 - \left(\frac{1}{2}\right)}$ $= -\frac{a}{2048}$	M1: Apply formula for sum to infinity from the 15th term onwards, including the 15th term A1: do not accept 2^{11}
	Alternatively Required sum $= S_{\infty} - (T_1 + T_2 + \dots + T_{14})$ $= \frac{-4a}{1 - \left(\frac{1}{2}\right)} - \frac{(-4a)\left(1 - \left(\frac{1}{2}\right)^{14}\right)}{1 - \left(\frac{1}{2}\right)}$ $= -8a + 8a\left(1 - \left(\frac{1}{2}\right)^{14}\right)$ $= 8a\left(1 - \left(\frac{1}{2}\right)^{14} - 1\right)$ $= -\frac{a}{2048}$	

Q2	Suggested Solutions	Marking Scheme
	$\sum_{r=1}^n \frac{1}{\sqrt{r} + \sqrt{r-1}}$ $= \sum_{r=1}^n \frac{\sqrt{r} - \sqrt{r-1}}{(\sqrt{r} + \sqrt{r-1})(\sqrt{r} - \sqrt{r-1})}$ $= \sum_{r=1}^n \frac{\sqrt{r} - \sqrt{r-1}}{r - (r-1)}$ $= \sum_{r=1}^n (\sqrt{r} - \sqrt{r-1})$ $= \begin{bmatrix} (\sqrt{1} - \sqrt{0}) \\ +(\sqrt{2} - \sqrt{1}) \\ +(\sqrt{3} - \sqrt{2}) \\ \vdots \\ +(\sqrt{n-1} - \sqrt{n-2}) \\ +(\sqrt{n} - \sqrt{n-1}) \end{bmatrix}$ $= \sqrt{n}$	M1: rationalize the denominator M1: method of differences A1: $\sqrt{n}$
	$u_{17} + \dots + u_{100} = \sqrt{100} - \sqrt{16} = 10 - 4 = 6$	M1: $S_{100} - S_{16}$ A1: 6

Q3	Suggested Solutions	Marking Scheme
	Volume of the hemisphere of radius r cm $= \frac{1}{2} \left(\frac{4}{3} \pi r^3 \right) = \frac{2}{3} \pi r^3$ Volume of hemisphere of radius $r/2$ cm $= \frac{1}{2} \left(\frac{4}{3} \pi \left(\frac{r}{2} \right)^3 \right) = \frac{1}{12} \pi r^3$ Volume of the rectangular cuboid = $2h^3$ Total volume of the model $= \frac{2}{3} \pi r^3 + \frac{1}{12} \pi r^3 + 2h^3$ $= \frac{3}{4} \pi r^3 + 2h^3$ Given the volume of the model is a fixed value 500 m^3, $\frac{3}{4} \pi r^3 + 2h^3 = 500$ $\Rightarrow 2h^3 = 500 - \frac{3}{4} \pi r^3$ $\Rightarrow h = \left(250 - \frac{3}{8} \pi r^3 \right)^{1/3}$ Surface area of hemisphere of radius r m = $\frac{1}{2} (4\pi r^2) = 2\pi r^2$ Surface area of hemisphere of radius $r/2$ m $= \frac{1}{2} \left(4\pi \left(\frac{r}{2} \right)^2 \right) = \frac{1}{2} \pi r^2$ Surface area of the walls and the roof $= 2(2h^2) + 2h^2 + 2h^2 - \pi r^2 - \frac{1}{4} \pi r^2$ $= 8h^2 - \frac{5}{4} \pi r^2$ Surface area of the floor = $2h^2$ Total surface area of the model, S $= 8h^2 - \frac{5}{4} \pi r^2 + 2h^2 + 2\pi r^2 + \frac{1}{2} \pi r^2$ $= 10h^2 + \frac{5}{4} \pi r^2$	M1: Find the volume of the model (allow consideration of sphere) M1: Equate the volume of the model to 500 and find h or h^3 in terms of r

Q3	Suggested Solutions	Marking Scheme												
	Substitute $h = \left(250 - \frac{3}{8}\pi r^3\right)^{1/3}$ in S: $S = 10\left(250 - \frac{3}{8}\pi r^3\right)^{2/3} + \frac{5}{4}\pi r^2$ Differentiating S with respect to r: $\begin{aligned}\frac{dS}{dr} &= 10\left(\frac{2}{3}\right)\left(-\frac{9}{8}\pi r^2\right)\left(250 - \frac{3}{8}\pi r^3\right)^{-1/3} + \frac{5}{2}\pi r \\ &= \left(-\frac{15}{2}\pi r^2\right)\left(250 - \frac{3}{8}\pi r^3\right)^{-1/3} + \frac{5}{2}\pi r\end{aligned}$ Equating $\frac{dS}{dr}$ to 0 and using GC to solve,  we have $r = \sqrt[3]{\frac{2000}{3\pi + 216}} = 2.070186387 = 2.07$ (3 s.f.) By 1st derivative test, <table><tr><td>r</td><td>2</td><td>$\sqrt[3]{\frac{2000}{3\pi + 216}}$</td><td>2.1</td></tr><tr><td>$\frac{dS}{dr}$</td><td>0.5541885</td><td>0</td><td>-0.248207</td></tr><tr><td>Slope</td><td>/</td><td>—</td><td>\</td></tr></table> $r = 2.07$ gives maximum S. Alternatively, by 2nd derivative test, By GC, $\left.\frac{d^2S}{dr^2}\right _{x=2.0702} = -8.1966725 > 0$ Since $\frac{d^2S}{dr^2} < 0$, $r = 2.07$ gives maximum S.	r	2	$\sqrt[3]{\frac{2000}{3\pi + 216}}$	2.1	$\frac{dS}{dr}$	0.5541885	0	-0.248207	Slope	/	—	\	M1: Obtain surface area of the model in terms of r (by substitution) M1: Find $\frac{dS}{dr}$ and equate to 0 A1: $r = \sqrt[3]{\frac{2000}{3\pi + 216}}$ or 2.07 M1: 1st derivative test with specific values or $\frac{d^2S}{dr^2} = -8.055131269$ A1: Correct conclusion
r	2	$\sqrt[3]{\frac{2000}{3\pi + 216}}$	2.1											
$\frac{dS}{dr}$	0.5541885	0	-0.248207											
Slope	/	—	\											

Q4	Suggested Solutions	Marking Scheme
(a)	$f'(x) = -k + \frac{9}{(x-1)^2}$ 	B1 : Equations of asymptotes and labelled B1 : x-intercepts with $x = 1 - \frac{3}{\sqrt{k}} < 0$ (accept $\sqrt{\frac{9}{k}}$, or do not use perfect square) B1 : shape of curve  (Allow 1 slip of asymptotic error)
(b)	 The graphs of $y = 10 - x - \frac{9}{x-1}$ and $y = \frac{9}{ x-1 }$ intersects at $x = p$ and $x = q$. Using G.C. to solve, we have $p = 4$ and $q = 7$. Hence for $f(x) \geq \frac{9}{ x-1 }$, $x < 1$ or $4 \leq x \leq 7$.	B1 : shape of $y = f(x)$ B1: Equations of asymptotes ($x = 1$, $y = 10 - x$) M1: Intersection of both graphs at $x = 4$ and $x = 7$ A1: $4 \leq x \leq 7$ A1: $x < 1$ (deduct 1 if use "and")

Q5	Suggested Solutions	Marking Scheme
(i)	Translate graph by b units in the positive x-direction Scale graph parallel to the x-axis by a factor of $\frac{1}{c}$. Scale graph parallel to the y-axis by a factor of a. Or Scale graph parallel to the x-axis by a factor of $\frac{1}{c}$. Translate graph by $\frac{b}{c}$ units in the positive x-direction. Scale graph parallel to the y-axis by a factor of a.	B3, 2, 1: complete and correct (based on the sequence) descriptions Note: Description must include (exclude) “units” for translation (scaling).
(ii)	$147 = a(-c-b)^2 = ac^2 + 2abc + ab^2 \quad \text{----- (1)}$ $3 = a(2c-b)^2 = 4ac^2 - 4abc + ab^2 \quad \text{----- (2)}$ $27 = a(4c-b)^2 = 16ac^2 - 8abc + ab^2 \quad \text{----- (3)}$ Using GC to solve the above equations, we get $ab^2 = 75, \quad abc = 30, \quad ac^2 = 12.$ Therefore $a = \frac{75}{b^2}, \quad c = \frac{30}{ab} = \frac{30}{\left(\frac{75}{b^2}\right)b} = \frac{2}{5}b$	M1: setting up 3 equations using the coordinates of the 3 points A1 B1
(iii)	$(-5, 25) \rightarrow (-5+b, 25) \rightarrow \left(\frac{-5+b}{c}, 25\right) \rightarrow \left(\frac{-5+b}{c}, 25a\right)$ Since $\frac{-5+b}{c} = 0$, we have $b = 5$. Hence $a = 3, \quad c = 2$	B1: $b = 5$ B1: $a = 3, \quad c = 2$ (ecf from ii)

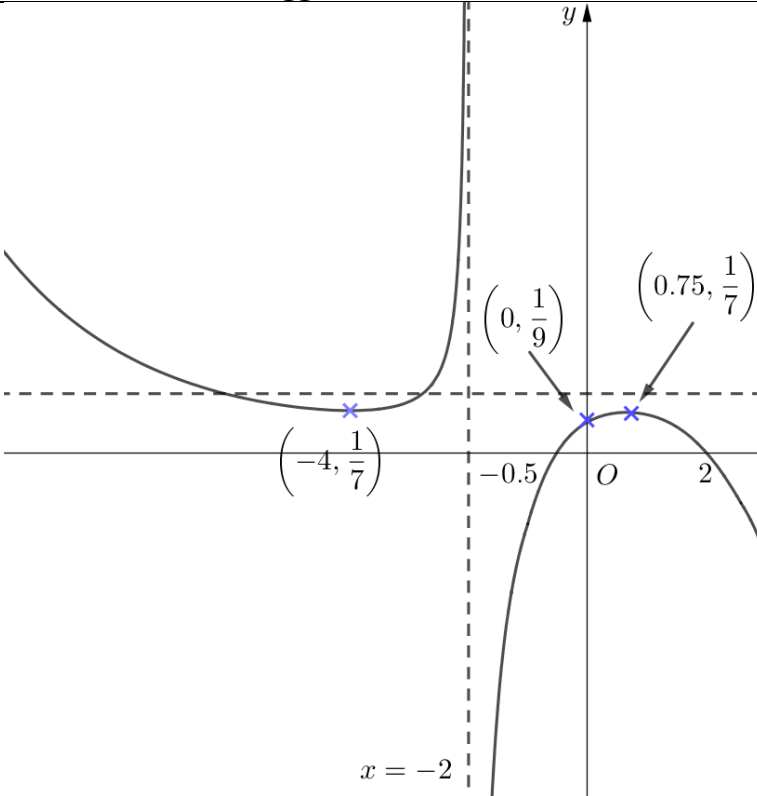
Q6	Suggested Solutions	Marking Scheme
(i)	For $x \in [1, \infty)$, $\frac{\pi}{4} \leq \tan^{-1} x < \frac{\pi}{2}$. Since $\tan^{-1} x \geq \frac{\pi}{4} > 0$ and $1+x^2 > 0$, $\frac{\tan^{-1} x}{1+x^2} > 0$. Thus, $\frac{\tan^{-1} x}{1+x^2}$ is positive. $\frac{d}{dx} \left(\frac{\tan^{-1} x}{1+x^2} \right) = \frac{(1+x^2) \left(\frac{1}{1+x^2} \right) - 2x \tan^{-1} x}{(1+x^2)^2}$ $= \frac{1-2x \tan^{-1} x}{(1+x^2)^2}$ Since $2x \geq 2$ and $\frac{\pi}{4} \leq \tan^{-1} x < \frac{\pi}{2}$, we have $2x \tan^{-1} x \geq \frac{\pi}{2} \approx 1.57.$ Thus, $1-2x \tan^{-1} x \leq 1 - \frac{\pi}{2} \approx -0.57 < 0$. Also, $(1+x^2)^2 > 0$ for all real values of x. This implies that $\frac{d}{dx} \left(\frac{\tan^{-1} x}{1+x^2} \right) = \frac{1-2x \tan^{-1} x}{(1+x^2)^2} < 0$. Hence, $\frac{\tan^{-1} x}{1+x^2}$ is decreasing.	B1: show numerator and denominator positive (can just see $\tan^{-1} x > 0$ for $x \in [1, \infty)$ or $\frac{\pi}{4} \leq \tan^{-1} x < \frac{\pi}{2}$ or allow graph of $\tan^{-1} x$ if asymptotes are seen clearly) *do not allow graph of $\frac{\tan^{-1} x}{1+x^2}$ B1: correct differentiation B1: correct conclusion for gradient less than 0
(b)	$\int_1^N \frac{\tan^{-1} x}{1+x^2} dx = \int_1^N \frac{1}{1+x^2} (\tan^{-1} x) dx$ $= \left[\frac{(\tan^{-1} x)^2}{2} \right]_1^N$ $= \frac{(\tan^{-1} N)^2}{2} - \frac{(\tan^{-1} 1)^2}{2}$ $= \frac{(\tan^{-1} N)^2}{2} - \frac{1}{2} \left(\frac{\pi}{4} \right)^2$ $= \frac{(\tan^{-1} N)^2}{2} - \frac{\pi^2}{32}$	M1: Using relevant standard formula to integrate M1: $\int_a^b f(x) dx = F(b) - F(a)$ B1: $\lim_{N \rightarrow \infty} (\tan^{-1} N) = \frac{\pi}{2}$ A1: $\frac{\pi^2}{8} - \frac{\pi^2}{32} = \frac{3\pi^2}{32}$ A1: conclude that it is “finite” thus converges

Q6	Suggested Solutions	Marking Scheme
	$\int_1^{\infty} \frac{\tan^{-1} x}{1+x^2} dx = \lim_{N \rightarrow \infty} \int_1^N \frac{\tan^{-1} x}{1+x^2} dx$ $= \lim_{N \rightarrow \infty} \left(\frac{(\tan^{-1} N)^2}{2} - \frac{\pi^2}{32} \right)$ $= \frac{1}{2} \left(\frac{\pi}{2} \right)^2 - \frac{\pi^2}{32}$ $= \frac{3\pi^2}{32},$ which is finite. Also, from (i), $\frac{\tan^{-1} x}{1+x^2}$ is non-negative and non-increasing. Hence, by the Maclaurin-Cauchy test, $\sum_{r=1}^{\infty} \frac{\tan^{-1} r}{1+r^2}$ converges.	P: $\left[\frac{(\tan^{-1} x)^2}{2} \right]_1^{\infty}, \tan^{-1} \infty$

Q7	Suggested Solutions	Marking Scheme
(a)	$\int_0^{\pi} \cos\left(\frac{x}{2}\right) \cos\left(\frac{x}{3}\right) + \pi dx$ $= \int_0^{\pi} \frac{1}{2} \cos\left(\frac{5x}{6}\right) + \frac{1}{2} \cos\left(\frac{x}{6}\right) + \pi dx$ $= \left[\frac{1}{2} \times \frac{6}{5} \sin\left(\frac{5x}{6}\right) + \frac{1}{2} \times 6 \sin\left(\frac{x}{6}\right) + \pi x \right]_0^{\pi}$ $= \frac{3}{5} \times \frac{1}{2} + 3 \times \frac{1}{2} + \pi^2$ $= \frac{9}{5} + \pi^2$	M1: factor formula A1
(b)(i)	Sum of area of $2n$ rectangles $= 2$ (Sum of area of n rectangles from $0 \leq x \leq \frac{\pi}{2}$) $= 2 \times \frac{\pi}{2n} \left\{ f\left(\frac{\pi}{2n}\right) + f\left(\frac{2\pi}{2n}\right) + \dots + f\left(\frac{n\pi}{2n}\right) \right\}$ $= \frac{\pi}{n} \left\{ 2 \cos^2\left(\frac{\pi}{2n}\right) + 1 + 2 \cos^2\left(\frac{2\pi}{2n}\right) + 1 + \dots + 2 \cos^2\left(\frac{(n-1)\pi}{2n}\right) + 1 + 2 \cos^2\left(\frac{n\pi}{2n}\right) + 1 \right\}$ $= \frac{\pi}{n} \left\{ 2 \cos^2\left(\frac{\pi}{2n}\right) + 2 \cos^2\left(\frac{2\pi}{2n}\right) + \dots + 2 \cos^2\left(\frac{(n-1)\pi}{2n}\right) + 2 \cos^2\left(\frac{n\pi}{2n}\right) + n \right\}$ $= \frac{2\pi}{n} \left\{ \cos^2\left(\frac{\pi}{2n}\right) + \cos^2\left(\frac{2\pi}{2n}\right) + \dots + \cos^2\left(\frac{(n-1)\pi}{2n}\right) + 0 \right\} + \frac{\pi}{n} \times n$ $= \frac{2\pi}{n} \left\{ \cos^2\left(\frac{\pi}{2n}\right) + \cos^2\left(\frac{2\pi}{2n}\right) + \dots + \cos^2\left(\frac{(n-1)\pi}{2n}\right) \right\} + \pi$	B1: sum of area of n rectangles of height $f\left(\frac{k\pi}{2n}\right)$ B1: summing all the '1's B1: $\cos\left(\frac{\pi}{2}\right) = 0$

Q7	Suggested Solutions	Marking Scheme
(b)(ii)	$S_{\infty} = \int_0^{\pi} 2 \cos^2(x) + 1 \, dx$ $= \int_0^{\pi} \cos 2x + 2 \, dx$ $= \left[\frac{\sin 2x}{2} + 2x \right]_0^{\pi}$ $= 2\pi$	B1: Relate to $\int_0^{\pi} 2 \cos^2(x) + 1 \, dx$ M1: double angle formula A1

Q8	Suggested Solutions	Marking Scheme
(i)	$\frac{dx}{dt} = k(x - 40)$ $\int \frac{1}{x - 40} dx = \int k dt$ $\ln x - 40 = kt + C$ $x - 40 = e^{kt+C} \quad (\because x \geq 40)$ $x = 40 + Ae^{kt}, \text{ where } A \text{ is a constant}$ When $t = 0, x = 60$, $A = 60 - 40 = 20$ When $t = 1800, x = 50$, $50 = 40 + 20e^{1800k} \Rightarrow k = \frac{1}{1800} \ln \frac{1}{2} = -\frac{1}{1800} \ln 2$ Thus, we have $x = 40 + 20e^{-\left(\frac{\ln 2}{1800}\right)t}$	B1: DE M1: variable separable (overall deduct 1 if no proper removal of modulus or invoking of conditions) M1: finding A using initial conditions M1: finding k A1
(ii)	Let y be the data downloaded by time T. $y = \int_0^T 40 + 20e^{\left(\frac{1}{1800} \ln \frac{1}{2}\right)t} dt$ $= \left[40t + \frac{20}{\frac{1}{1800} \ln \frac{1}{2}} e^{\left(\frac{1}{1800} \ln \frac{1}{2}\right)t} \right]_0^T$ $= 40T + \frac{36000}{\ln 0.5} e^{\left(\frac{1}{1800} \ln \frac{1}{2}\right)T} - \frac{36000}{\ln 0.5}$ Converting T to hours, and equating y to 500000, we have $40(3600T) + \frac{36000}{\ln 0.5} e^{\left(\frac{1}{1800} \ln \frac{1}{2}\right)(3600T)} - \frac{36000}{\ln 0.5} = 500000.$ Using GC, we have $T = 3.1163446$. Thus, the user can stream for 3.1 hours continuously.	M1: integrating x with initial conditions M1: converting t and set up equation A1: 3.1 (requires units)

Q9	Suggested Solutions	Marking Scheme
(i)		$y = \frac{1}{f(x)}$
	B1: correct left portion with correct minimum turning point and vertical asymptote B1: correct middle portion with correct maximum turning point and axial intercepts B1: correct right end portion with horizontal asymptote and vertical asymptote B1: turning points are at the same y-value (relative position) B1: eqn of asymptotes B1: all axial intercepts B1: all turning points B1: shape with asymptotic behaviour Max of 2 MS	
(ii)	For the inverse of f to exist, smallest $k = 0.75$	B1: smallest $k = 0.75$
(iii)	$g: x \mapsto \frac{2x+a}{x-b}$ for $x \in \mathbb{R}, x > b, a, b \in \mathbb{R}^+$. The graph of $y = \frac{2x+a}{x-b} = 2 + \frac{a+2b}{x-b}$ has vertical asymptote $x = b$ and horizontal asymptote $y = 2$. For the domain $x > b$, $R_g = (2, \infty)$.	M1: long division and attempt to find asymptote A1: $R_g = (2, \infty)$

Q9	Suggested Solutions	Marking Scheme
(iv)	Let $(fg)^{-1}(0) = x$, from the mapping above, $fg(x) = 0$, which implies $g(x) = f^{-1}(0)$. From the graph of f in the domain $[0.75, \infty)$, $f^{-1}(0) = 4$. Solve $g(x) = 4$ $\frac{2x+a}{x-b} = 4$ $\Rightarrow 2x+a = 4x-4b$ $\Rightarrow 2x = a+4b$ $\Rightarrow x = 0.5a+2b$	B1: $f^{-1}(0) = 4$ M1: Solve $g(x) = 4$ A1: $x = 0.5a + 2b$

Q10	Suggested Solutions	Marking Scheme
(i)	$x = 0$ (or y -axis)	B1
(ii)(a)	Volume $= \pi \int_0^{e-1} \frac{\ln(y+1)}{(y+1)^2} dy$ $= \pi \int_0^{e-1} \frac{1}{(y+1)^2} \ln(y+1) dy$ $= \pi \left[\frac{-1}{(y+1)} \ln(y+1) \right]_0^{e-1} - \pi \int_0^{e-1} -\frac{1}{(y+1)} \frac{1}{(y+1)} dy$ $= \pi \left(-\frac{1}{e} \right) + \pi \int_0^{e-1} \frac{1}{(y+1)^2} dy$ $= \pi \left(-\frac{1}{e} \right) + \pi \left[\frac{-1}{(y+1)} \right]_0^{e-1}$ $= \pi \left(-\frac{1}{e} \right) + \pi \left(-\frac{1}{e} + 1 \right)$ $= \pi \left(1 - \frac{2}{e} \right)$	M1: volume formula with correct limits M1: applying by parts M1: applying definite integral A1
(ii)(b)	$x^2 = \frac{\ln(y+1)}{(y+1)^2}$ $x^2 (y+1)^2 = \ln(y+1)$ $x^2 \left[2(y+1) \frac{dy}{dx} \right] + 2x(y+1)^2 = \frac{1}{y+1} \frac{dy}{dx}$ When $y = e-1$, $x^2 = \frac{\ln(e)}{(e)^2} = \frac{1}{e^2} \Rightarrow x = \pm \frac{1}{e}$ When $x = \frac{1}{e}$, $y = e-1$, we have $\frac{1}{e^2} [2e] \frac{dy}{dx} + \frac{2}{e} (e)^2 = \frac{1}{e} \frac{dy}{dx}$ $\frac{1}{e} \frac{dy}{dx} = -2e$ $\frac{dy}{dx} = -2e^2$ Thus, the equation of normal is $y - (e-1) = \frac{1}{2e^2} \left(x - \frac{1}{e} \right).$ Since C is symmetrical about the y-axis, and P and Q share the same y-coordinate, their normals must meet at the y-axis.	M1: applying differentiation correctly M1: finding gradient at point M1: equation of normal

Q10	Suggested Solutions	Marking Scheme
	When $x = 0$, $y - (e - 1) = \frac{1}{2e^2} \left(-\frac{1}{e} \right)$ $y = e - \frac{1}{2e^3} - 1$ $\therefore N \left(0, e - \frac{1}{2e^3} - 1 \right)$	M1: deduction about $x = 0$ or intersecting both normals A1

Q11	Suggested Solutions	Marking Scheme
(a)	$w - 6\alpha z = -5\alpha i + 3$ --- (1) $\alpha z^2 + iw = 10\alpha + 3i$ --- (2) Multiply (1) by i: $iw - 6\alpha iz = 5\alpha + 3i$ --- (3) By elimination method, taking (2) – (3) : $\alpha z^2 + 6\alpha iz = 5\alpha$ $\Rightarrow z^2 + 6iz - 5 = 0$ $z = \frac{-6i \pm \sqrt{(6i)^2 - 4(1)(-5)}}{2(1)}$ $= \frac{-6i \pm \sqrt{-16}}{2}$ $= \frac{-6i \pm 4i}{2}$ $= \frac{-10i}{2} \text{ or } \frac{-2i}{2}$ $\therefore z = -5i$ or $-i$ (rejected since $z > 1$) Substitute $z = -5i$ into (1): $w = 6\alpha(-5i) - 5\alpha i + 3 = 3 - 35\alpha i$.	M1: Apply elimination method (accept substitution method too) to reach a quadratic equation in z M1: Apply quadratic formula to solve $z^2 + 6iz - 5 = 0$. (Accept method by letting $z = x + iy$) A1: $z = -5i$ and $w = 3 - 35\alpha i$
(b)(i)	$\arg z_1 = -\tan^{-1} \frac{\sqrt{3}}{3} = -\frac{\pi}{6}$ and $\arg z_2 = \frac{\pi}{2}$ $\arg \left(\frac{z_1^3}{z_2^2} \right)^n = n \arg \left(\frac{z_1^3}{z_2^2} \right)$ $= n(\arg z_1^3 - \arg z_2^2)$ $= n(3 \arg z_1 - 2 \arg z_2)$ $= n \left[3 \left(-\frac{\pi}{6} \right) - 2 \left(\frac{\pi}{2} \right) \right]$ $= -\frac{3\pi}{2} n$ $\arg \left(\frac{z_1^3}{z_2^2} \right)^n = \frac{\pi}{2} n \text{ (in principal range)}$ For $\frac{z_1^{3n}}{z_2^{2n}} = \left(\frac{z_1^3}{z_2^2} \right)^n$ to be purely imaginary with a positive imaginary part, $\arg \left(\frac{z_1^3}{z_2^2} \right)^n = \dots, -\frac{3\pi}{2}, \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \dots = (4k+1)\frac{\pi}{2}, k \in \mathbb{Z}$	M1: apply properties of argument to find argument of $\left(\frac{z_1^3}{z_2^2} \right)^n$

Q11	Suggested Solutions	Marking Scheme
	$\Rightarrow \frac{\pi}{2}n = (4k+1)\frac{\pi}{2}$ $\Rightarrow n = 4k+1$ For integer $n = \dots, -3, 1, 5, 9, 13, \dots = 4k+1, k \in \mathbb{Z}$	M1: State at least 2 correct values of $\arg(\text{purely positive imaginary numbers})$ or give general form A1: $n = 4k+1, k \in \mathbb{Z}$ or $n = 4k-3, k \in \mathbb{Z}$ Must see $k \in \mathbb{Z}$ or see at least 3 values with ...
(b)(ii)	$z_1^2 = 9 - 3 - i(6\sqrt{3}) = 6 - i(6\sqrt{3})$ $z_1^3 = 27 - i(27\sqrt{3}) - 27 + (3\sqrt{3})i = -i(24\sqrt{3})$ Since $z_1 = 3 - i\sqrt{3}$ is a root of $px^3 - 9x^2 + 6x + 36 = 0$, $p[-(24\sqrt{3})i] - 9[6 - (6\sqrt{3})i] + 6(3 - i\sqrt{3}) + 36 = 0$ $p[-(24\sqrt{3})i] = (48\sqrt{3})i$ $p = 2$	M1: Substitute w_1 into equation to solve for p (must see the working process to solve for p) A1: $p = 2$
	Since the coefficients of the polynomial equation are real, by conjugate theorem, $z_1 = 3 + (\sqrt{3})i$ is one of the roots. Thus, the remaining root is a real root. Let the remaining factor be $2x + a$. We have $2x^3 - 9x^2 + 6x + 36 = (2x + a)(x - 3 - (\sqrt{3})i)(x - 3 + (\sqrt{3})i)$ $= (2x + a)(x^2 - 6x + 12)$ By observation, $a = 3$. $2x^3 - 9x^2 + 6x + 36 = (2x + 3)(x - 3 - (\sqrt{3})i)(x - 3 + (\sqrt{3})i)$ Note: $2x^3 - 9x^2 + 6x + 36 \neq \left(x + \frac{3}{2}\right)(x - 3 - (\sqrt{3})i)(x - 3 + (\sqrt{3})i) \text{ as } \left(x + \frac{3}{2}\right) \text{ is not accepted.}$	M1: Apply conjugate theorem A1: $2x^3 - 9x^2 + 6x + 36 = (2x + 3)\left(x - e^{i\left(-\frac{\pi}{6}\right)}\right)\left(x - e^{i\left(\frac{\pi}{6}\right)}\right)$
	Alternatively, Since the coefficients of the polynomial equation are real, by conjugate theorem, $z_1 = 3 + (\sqrt{3})i$ is one of the roots. Thus, the remaining root is a real root. $px^3 - 9x^2 + 6x + 36 = (Ax + B)(x - 3 - (\sqrt{3})i)(x - 3 + (\sqrt{3})i)$ $\Rightarrow px^3 - 9x^2 + 6x + 36 = (Ax + B)(x^2 - 6x + 12)$	

Q11	Suggested Solutions	Marking Scheme
	By comparing coefficient of x^3: $p = A$ constant term : $36 = 12B \Rightarrow B = 3$ x^2 : $-9 = -6A + B \Rightarrow A = 2$ $\therefore p = 2$ $\therefore 2x^3 - 9x^2 + 6x + 36 = (2x + 3)(x - 3 - (\sqrt{3})i)(x - 3 + (\sqrt{3})i)$	

Q12	Suggested Solutions	Marking Scheme
(i)	$\overrightarrow{OA} = \begin{pmatrix} 0 \\ 4 \\ -8 \end{pmatrix} \quad \overrightarrow{OB} = \begin{pmatrix} 8 \\ 0 \\ -4 \end{pmatrix} \quad \overrightarrow{OC} = \begin{pmatrix} 4 \\ 8 \\ 0 \end{pmatrix}$ $\overrightarrow{AB} = \begin{pmatrix} 8 \\ -4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \overrightarrow{AC} = \begin{pmatrix} 4 \\ 4 \\ 8 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ Normal of plane $\pi = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 3 \end{pmatrix} = -3 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ Equation of π : $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ -8 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 12$ $x + y - z = 12$	M1: Attempt to find coordinates of A, B, C. M1: Find any 2 vectors parallel to plane ABC. M1: Find normal A1
(ii)	Equation of line consisting of O and F $\mathbf{r} = \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ For coordinates of F, $\lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 12$ $\lambda = 4$ $\overrightarrow{OF} = 4 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ Coord of F is $(4, 4, -4)$.	M1: Find eqn of line OF M1: Find point of intersection. A1: in coordinates
(iii)	Equation of line DE: $\mathbf{r} = \begin{pmatrix} 8 \\ 0 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ Let the point on DE nearest to P be $\begin{pmatrix} 8 + \lambda \\ 0 \\ 4 \end{pmatrix}$.	M1: representing point using equation of line

Q12	Suggested Solutions	Marking Scheme
	$\left(\begin{pmatrix} 8+\lambda \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 11 \\ 1 \end{pmatrix} \right) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$ $6 + \lambda = 0$ $\lambda = -6$ Hence, $\overrightarrow{OQ} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}$. Shortest distance from P to line $ED = PQ$ $= \left \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 11 \\ 1 \end{pmatrix} \right = \left \begin{pmatrix} 0 \\ -11 \\ 3 \end{pmatrix} \right $ $= \sqrt{130} \text{ m}$ The shortest distance the pigeon has to fly is $\sqrt{130} \text{ m}$.	M1: perpendicular A1 B1: or 11.4 m
(iv)	$\frac{x-1}{3} = \frac{2-y}{4} = \frac{z+3}{1} \Rightarrow \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ $\mathbf{r} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -11 \\ 3 \end{pmatrix}$ Equating the two flight paths' equations, $\begin{pmatrix} 1+3\mu \\ 2-4\mu \\ -3+\mu \end{pmatrix} = \begin{pmatrix} 2 \\ -11\lambda \\ 4+3\lambda \end{pmatrix}$ Solving the first 2 equations, we have $\mu = \frac{1}{3} \text{ and } \lambda = -\frac{2}{33}.$ However, substituting in the third equation, we have $-3 + \frac{1}{3} = -\frac{8}{3} \neq \frac{42}{11} = 4 - \frac{2}{11}.$ Thus, the flight paths do not cross.	M1: equation of either line M1: solving for unknowns A1: checking 3rd equation